Mohamed Affan Dhankwala

affandhankwala@gmail.com • github.com/affandhankwala • linkedin.com/in/affandhankwala

Professional Experience

Software Engineer - Raytheon Sept 2023 - present

- Improved performance, robustness, and maintainability by modernizing legacy aerospace PATRIOT missile-borne computer to align with modern architecture in the form of new software designs and FPGA-based hardware to meet customer needs.
- Enabled parallel software development in hardware-limited environments by simulating firmware interactions with memory-mapped I/O on Linux-based systems, allowing developers to test bare-metal register transactions and device behavior without physical hardware.
- Partnered with firmware, FPGA (configurable logic), quality and safety teams to deliver and validate safety-critical device drivers and memory maps for embedded hardware transactions on Polarfire SOC supporting aerospace applications.
- Developed, integrated, and debugged C/C++ software and Linux-based drivers on target hardware; delivering a successful stakeholder demo that secured approval for further development.
- Supported the algorithm team in developing modern aerospace autopilot algorithms, implementing FPGA based unit tests, and maintaining CMake build configurations to enable requirements verification and strengthening CI/CD efforts.
- Unblocked development efforts constrained by limited hardware availability by creating a Python script to convert IOX memory maps to DDR-backed memory.
- Owned the downlink component end-to-end, designing the architecture, defining stories, coordinating team execution, and validating PRs to meet stakeholder requirements and support high-fidelity demos.
- Built and maintained CI/CD pipelines to enforce code quality and design correctness.

Software Engineer Intern - Microsoft May 2018 - August 2018

- Designed, developed, and demoed a Kendo UI based video and text highlighting plugin for Massive open online courses (MOOCs) on the Worldwide Learning Team; enhancing content interaction and learner engagement.
- Conducted regular user interviews to identify pain points and iteratively refine solutions, improving usability and adoptions.
- Designed and implemented global and local databases to allow collaborative interaction on user annotations, improving the learning experience for MOOCs.
- Delivered a fully documented solution with 95% unit test coverage, reviewed and demoed to senior engineers and management.
- Participated in agile sprints, driving software development from planning to deployment.

Master's Projects

Textwave - RAG System January 2025 - February 2025

- Processed and encoded large document corpus with multiple chunking strategies and training FAISS sequential re-rankers to enhance retrieval quality.
- Leveraged a pretrained Mistral AI LLM to generate contextually grounded answers for natural language queries.
- Performed offline evaluation and error analysis to verify model accuracy, grounding and robustness.

Techtrack - Real-Time Image/Video Recognition Service February 2025 - March 2025

- Built a multi-threaded AI video streaming service for real-time object detection and classification.
- Leverage YOLO to visualize bounding boxes for multiple objects and people while minimizing latency.
- Optimized processing pipelines for high-performance, low-latency interference.

Bionic Arm March 2023 - May 2023

- Built a distributed embedded system using dual Arduino boards communicating over Bluetooth to enable real-time motion mirroring.
- Developed firmware to process flex sensor inputs from a wearable 3D-printed glove to control motor actuated joints on an originally designed 3D printed robotic claw.
- Debugged timing, signal noise, and calibration issues to achieve stable, repeatable movement under continuous operation and demoed final results to University faculty staff.

Reinforcement Learning Environments and Agents January 2024 - December 2025

- Implemented custom Q-learning and SARSA agents in Python across multiple environments, including grid-based navigation, continuous control, and turn-based scenarios.
- Designed configurable state representations, reward functions, and exploration strategies using tabular methods.
- Evaluated learning dynamics using episode reward trends and policy convergence analysis, comparing performance across environments and algorithmic variants.

Education

MS in Artificial Intelligence, Johns Hopkins University December 2025
BS in Computer Engineering, Minor in Mathematics, University of Washington June 2023

Research Papers

M. A. Dhankwala, "A comparative study on hybrid genetic algorithm-boids as evolutionary algorithms" 2025.
M. A. Dhankwala, "Comparison of hierarchical navigable small world querying variants" 2025.

Languages, Tools, and Practices

- **Languages:** C++, C, C#, Python, Java, JavaScript, Bash
- **Tools:** Git, Azure DevOps, Linux, CMake, VXWorks, PyTest, Pytorch, Pandas, Scikit, Docker, EC2, DOORS, GDB
- **Practices:** Unit & Integration Testing, Requirements traceability, CI/CD, Model Eval & Benchmarking